Probing the Higgs Potential at 33 TeV

Prospects for measuring the trilinear coupling via $hh \to b\bar{b}\gamma\gamma$ at a High-Energy LHC

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Outline

Introduction and Motivation
- The Standard Model and the Higgs
- The Electroweak Phase Transition and λ_3
- New Physics in the Higgs Potential

Current Status of Di-Higgs Production
- Theoretical Status
- Di-Higgs Decay Channels
- Prospects at the HL-LHC and 100 TeV

Measuring λ_3 at a 33 TeV Collider
- Simulation of Di-Higgs Signal and Backgrounds
- Kinematic Analysis and Cuts
- Preliminary Results

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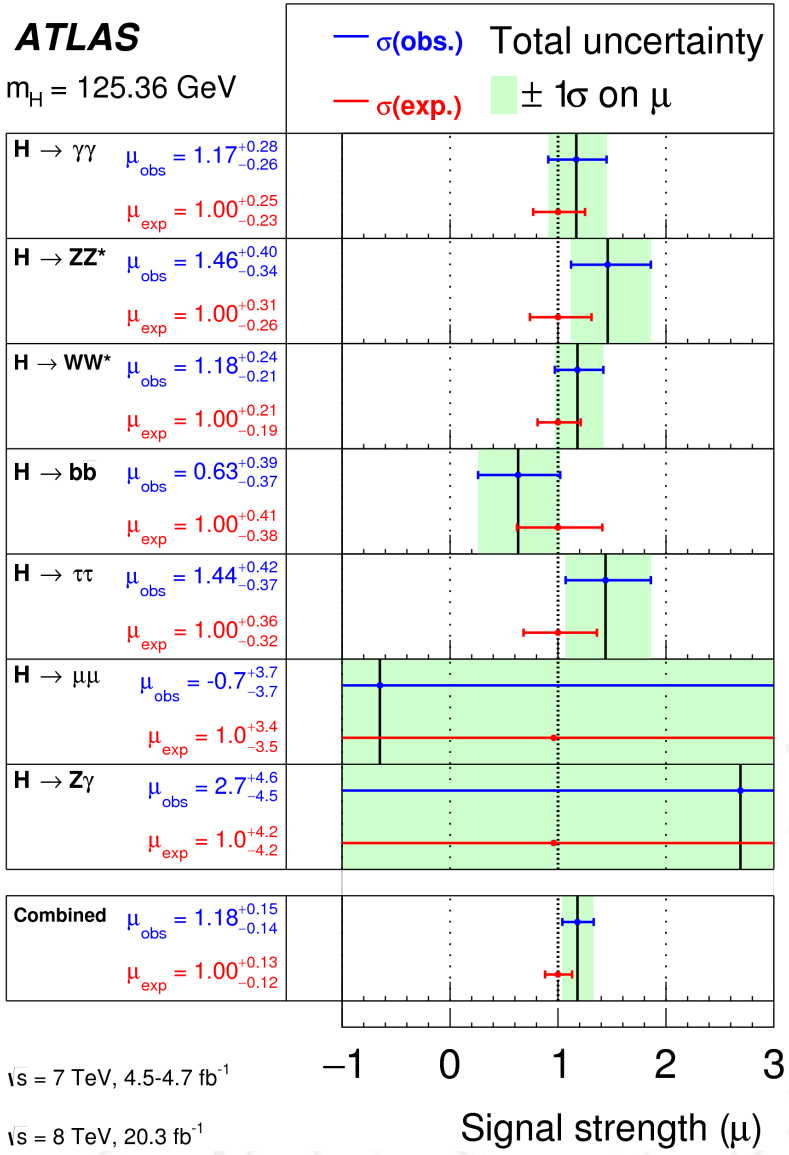
The Standard Model Higgs

The discovery of a Higgs-like particle "completes" the Standard Model

We anticipated *something* at the TeV scale to solve problems such as unitarity, to mediate spontaneous symmetry breaking, etc.

The SM Higgs is the only elementary scalar – unique and problematic!



Eur. Phys. J. C76 (2016) 6

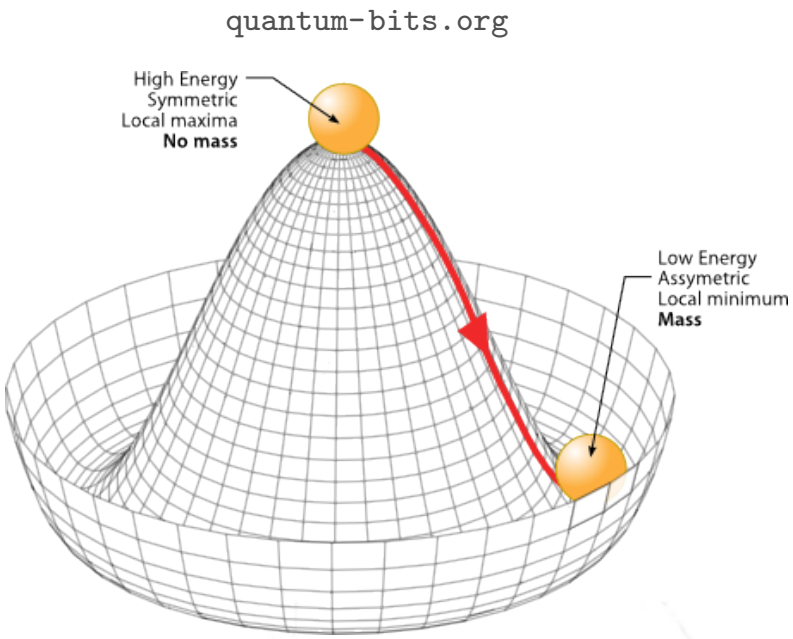
Spontaneous Symmetry Breaking

Canonical Higgs Potential:

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2$$

where

$$\phi = \frac{1}{\sqrt{2}} e^{i\xi^a T'_a} \begin{pmatrix} 0 \\ v + h \end{pmatrix}$$



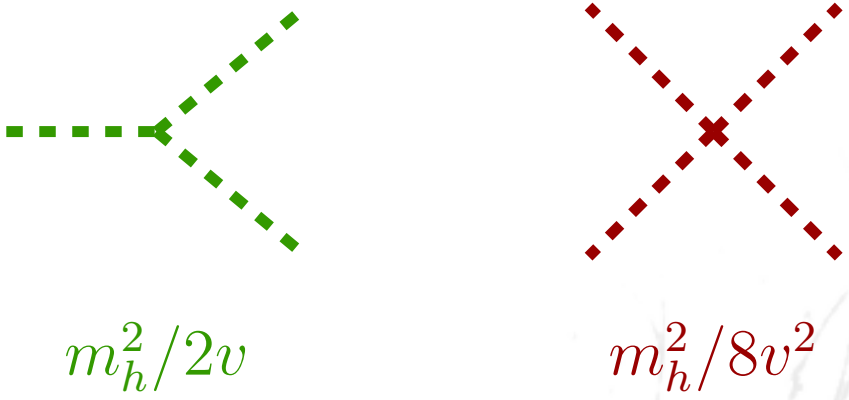
Unitarity and gauge symmetry fix form of h couplings to fermions, gauge bosons.

Scalar Interactions

After spontaneous symmetry breaking, the Higgs self-interactions take the form

$$\mathcal{L} \supset -\frac{1}{2}m_h^2 h^2 - \lambda_3 \frac{m_h^2}{2v} h^3 - \lambda_4 \frac{m_h^2}{8v^2} h^4$$

$\lambda_3 = \lambda_4 = 1$ in the Standard Model.



Fixed by known quantities! $m_h \simeq 125$ GeV, $v = 246$ GeV.

Importance of Scalar Self-Couplings

The Higgs is the only particle we know that couples directly to itself!

Gauge interactions change quantum numbers (color, helicity..)

Important to understand if this interaction exists and what it looks like.

Possible deviations could point to substructure of the Higgs, extended sectors, new physics..

The scalar potential doesn't need to be canonical form – can take any shape with first and second derivatives fixed by v and m_h.

The Electroweak Phase Transition (EWPT)

In the early universe interactions with the plasma give the Higgs an effective potential:

$$V_{\text{eff}}(h,T) \approxeq V_0(h) + V_{CW}(h) + V_T(h,T)$$

Tree-Level

Coleman-Weinberg Terms

Thermal Terms

The behavior of this potential could have profound effects on cosmology at $T \sim \mathcal{O}(100\text{ GeV})$.

By contrast, the CMB is at $T \sim 1$ eV and BBN is sensitive to $T \sim 1$ MeV.

The Standard Model EWPT

In the SM at high temperatures, V_{eff} takes the approximate form

$$V_{\text{eff}}(h,T) \approxeq \frac{1}{2}c(T^2 - T_c^2)h^2 - \frac{eT}{12\pi}\left(h^2\right)^{3/2} + \frac{\lambda}{4}h^4$$

where

$$T_c = \frac{m_h^2}{2c}, \qquad c = \frac{4m_t^2 + 4m_W^2 + 2m_Z^2 + m_H^2}{8v^2}$$

$$e = \frac{6m_W^3 + 3m_Z^3}{v^3}$$

Note that at $T > T_c$, the mass term is positive: $SU(2)_L \times U(1)_Y$ is unbroken.

We therefore expect a phase transition at high temperatures in the early universe.

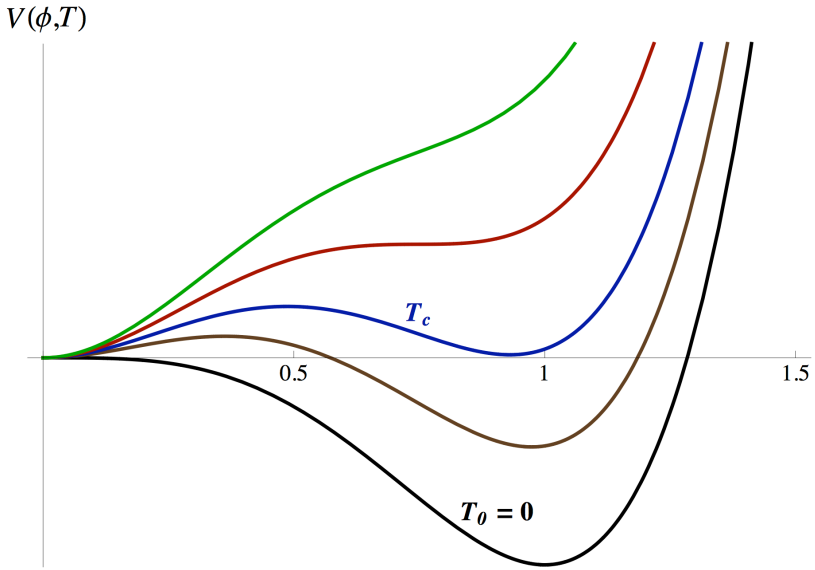
The Standard Model EWPT

If there is no barrier between the two minima thermal fluctuations drive a smooth *second order* phase transition.

This is the case in the standard model with $m_h = 125$ GeV.

An h^3 term creates a barrier between the two phases, and the phase transition instead proceeds via tunneling.

Such a phase transition is *first order*.



arXiv:1408.3533

Strong First Order Phase Transitions

The strength of the phase transition is characterized by the order parameter, $v(T)/T$.

If $v(T_c)/T_c \gtrsim 1$, we say the phase transition is *strongly first order*.

With our previous expansion, we see that

$\sim h^3$ term

$$\frac{v(T_c)}{T_c} \cong \frac{(e/4\pi)}{\lambda}$$

Fixed to SM value

which is > 1 only for $m_h \lesssim 48$ GeV.

To get a strong first order EWPT, we need additional contributions to the Higgs potential – these can be explored via colliders!

Collider Tests of the EWPT

To get a strong barrier, BSM physics coupled to the EW sector is necessary, which can be explored experimentally.

A characteristic scenario is extension via a singlet scalar, S, with or without mixing with the Higgs.

The EWPT can then be one-step or multi-step, via local minima of the additional scalar field.

For regions where direct production of S is unfeasible, deviations of $\lambda_3 \sim 10\%$ are necessary to produce a strong first order EWPT.

Model Classes of the EWPT

Chung, Long, Wang, arXiv:1209.1819

Chung et. al categorize models of a first order phase transition according to source of barrier term:

- Thermally Driven ($\sim -T(h^2)^{3/2}$)
  Arises from thermal loops over bosonic zero modes.
- Tree-Level Driven ($\sim h^3$)
  Additional scalar fields acquiring a vev can induce h^3 term in potential. Also arise from non-renormalizable operators e.g., h^6.
- Loop driven ($\sim h^4 \log h^2$)
  Coleman-Weinberg-like terms can generate interesting effects in non-standard potnetials.

New Physics in the Higgs Potential

An anomalous value of λ_3 could arise from new physics at high scales or from hidden sectors near the electroweak scale.

Gupta et. al (1305.6397) examined the maximal deviations in λ_3 allowed in a variety of models.

Assuming no additional states are accessible at the LHC.

Process	$\Delta\lambda_3$
Mixed-in Singlet	-18%
Composite Higgs	$\mathcal{O}(10\%)$
MSSM	-2% to -15%
NMSSM	-25%

Higher-Dimensional Operators

If new physics is at a high scale, we can parameterize the effects viewing the SM as an effective field theory.

At next order (dimension-6), there are 4 new operators involving ϕ that modify the Higgs self-interactions:

Violates custodial symmetry

$$\mathcal{O}_1 = \left(D^\mu \phi\right)^\dagger \phi \phi^\dagger \left(D_\mu \phi\right),$$

$$\mathcal{O}_2 = \frac{1}{2} \partial^\mu \left(\phi^\dagger \phi\right) \partial_\mu \left(\phi^\dagger \phi\right),$$

$$\mathcal{O}_3 = \frac{1}{3} \left(\phi^\dagger \phi\right)^3,$$

$$\mathcal{O}_4 = \left(D^\mu \phi\right)^\dagger \left(D_\mu \phi\right) \left(\phi^\dagger \phi\right).$$

Redundant via E.O.M.

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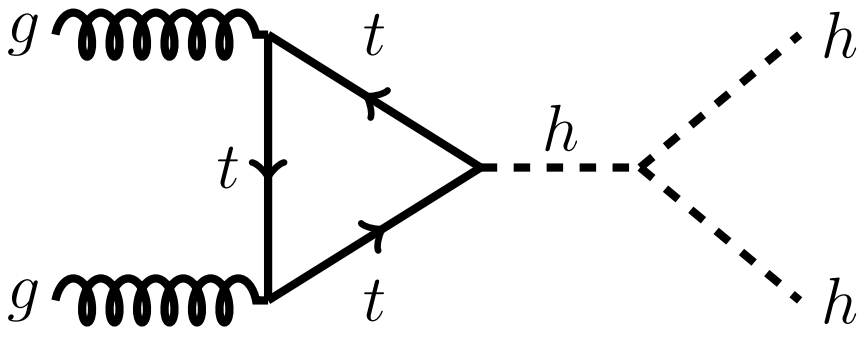
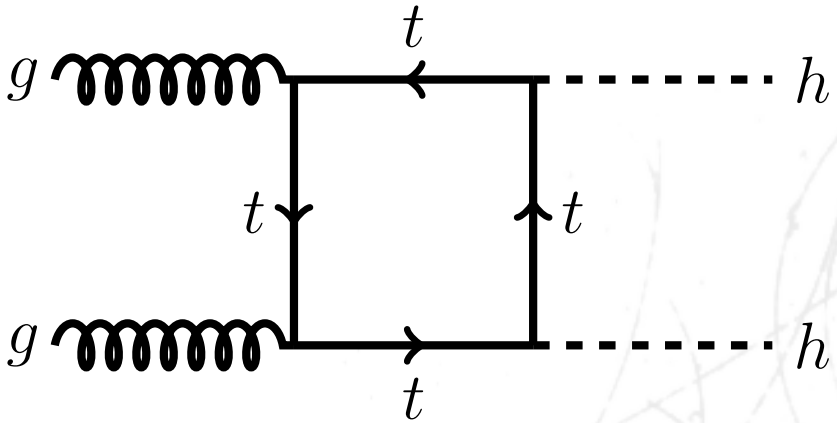
Di-Higgs Production as a Probe of λ_3

The λ_3 vertex contributes to di-Higgs production:



There's also an interference term:



Di-Higgs Production as a Probe of λ_3

The diagrams have different analytic behavior:

Suppression at high $\hat{s}$ due to off-shell Higgs!

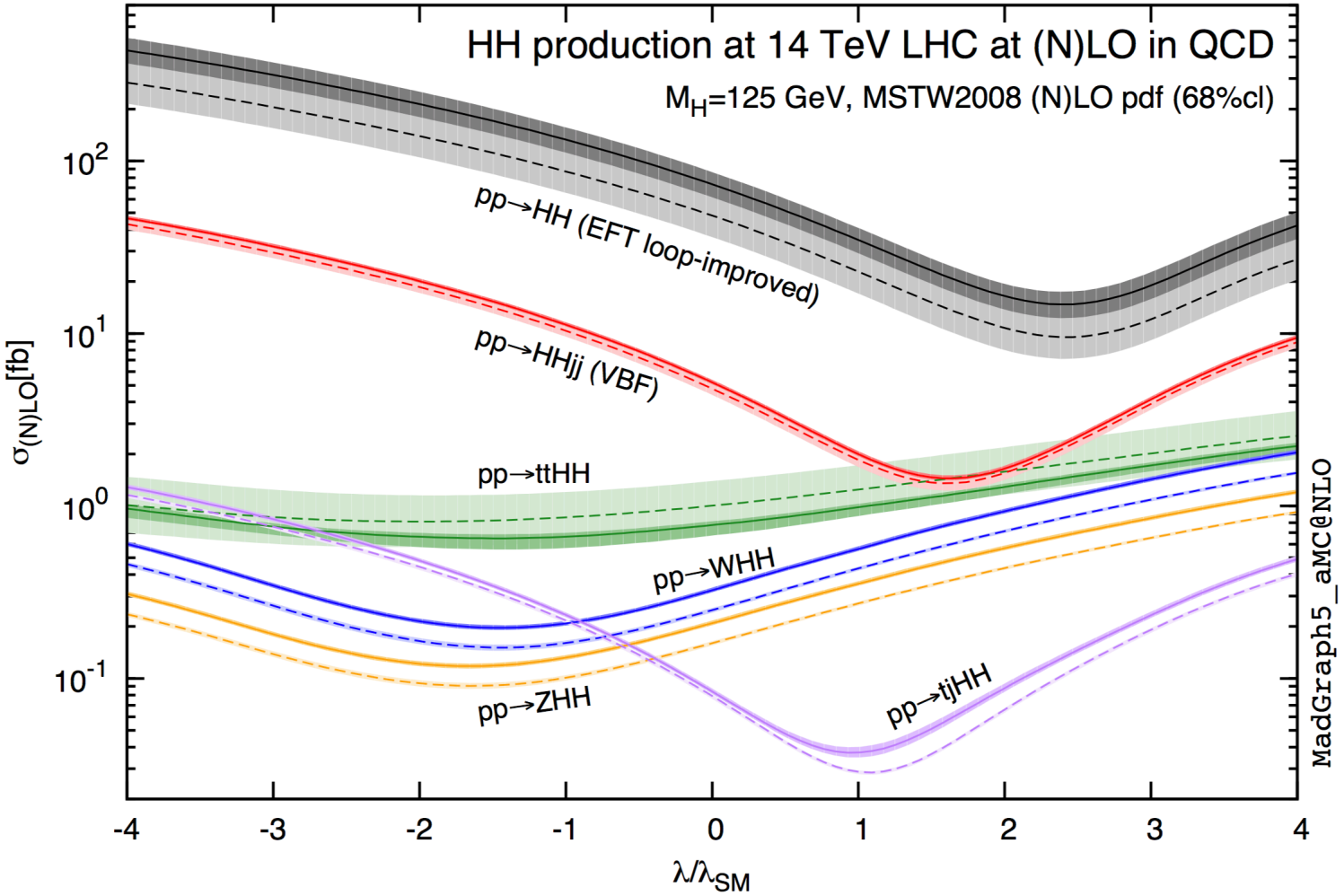
$$\mathcal{A}_\triangle \sim \lambda_3 \frac{\alpha_S}{4\pi} y_t \frac{m_h^2}{\hat{s}} \left(\log \frac{m_t^2}{\hat{s}} + i\pi \right)^2$$

$$\mathcal{A}_\square \sim -\frac{\alpha_S}{4\pi} y_t^2$$

Large cancellation between diagrams reduces sensitivity to λ_3.

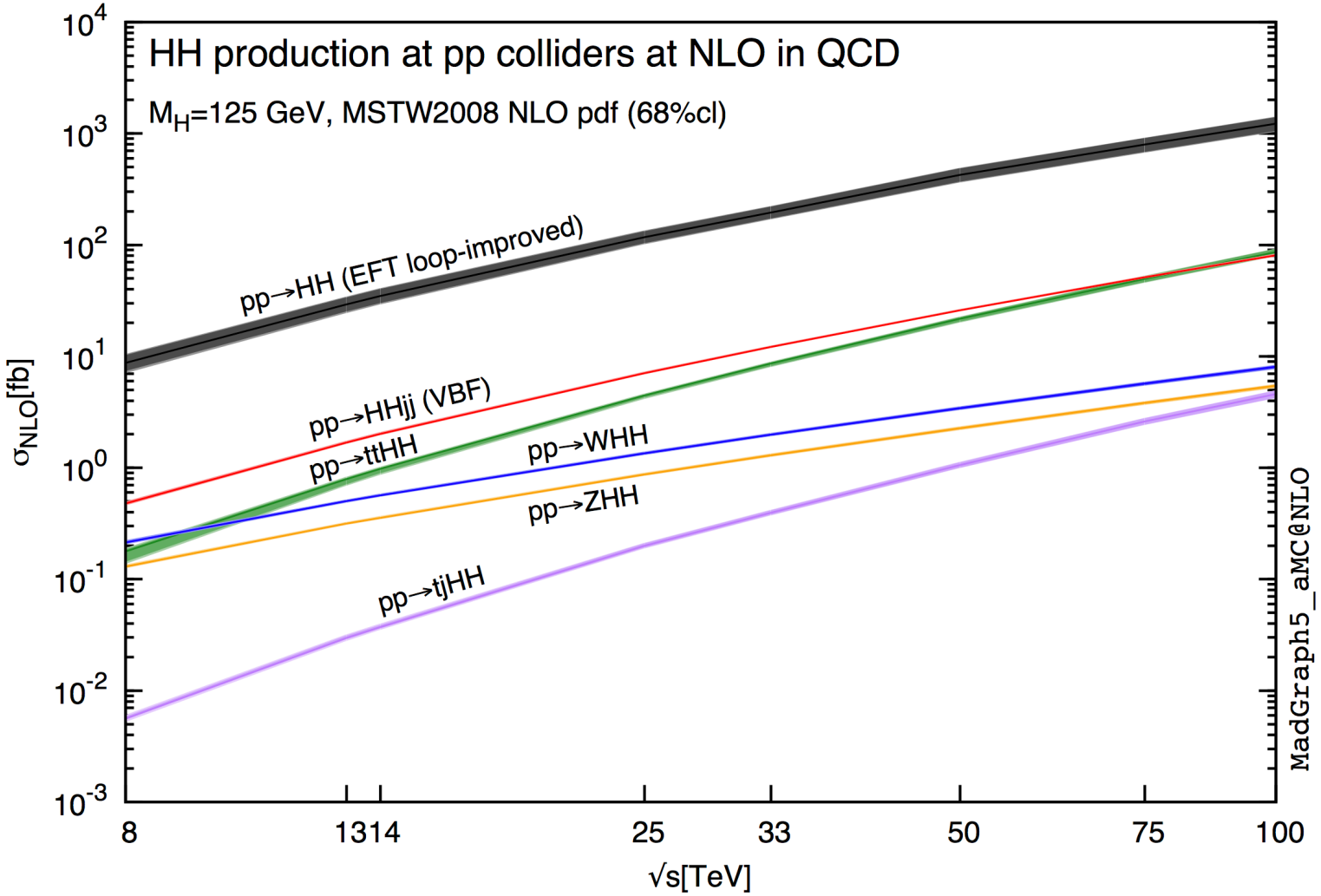
Production very sensitive to anomalous top couplings!

Dependence on λ_3



arXiv:1401.7340

Di-Higgs Production Cross Section



$\sigma_{gg\to hh}(14\text{ TeV}) = 44.5\text{ fb}$

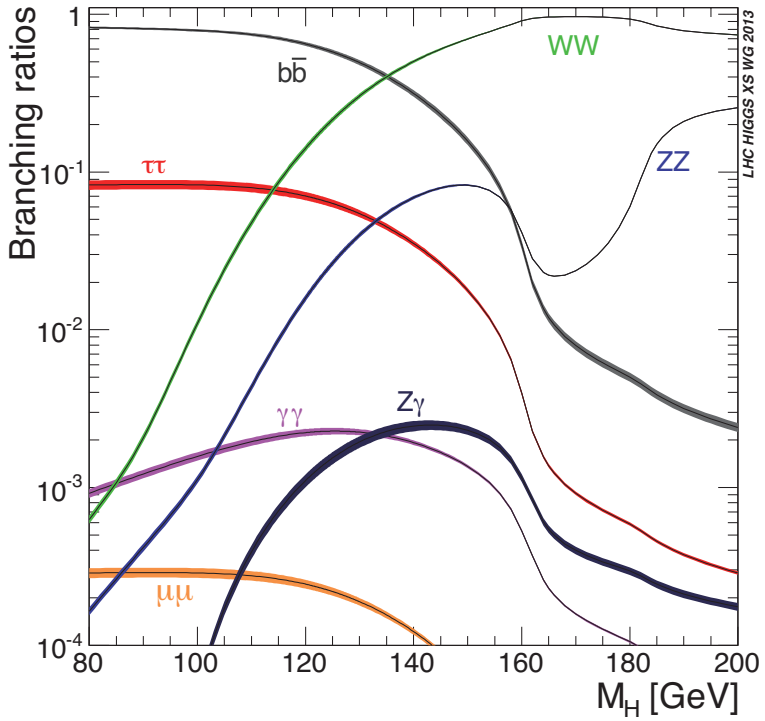
$\sigma_{gg\to hh}(33\text{ TeV}) = 265\text{ fb}$

$\sigma_{gg\to hh}(100\text{ TeV}) = 1762\text{ fb}$

Higgs Decay Modes

Final State	Branching Ratio
$b\bar{b}$	0.58
WW^*	0.215
gg	0.09
$\tau^+\tau^-$	0.06
ZZ^*	0.026
$\gamma\gamma$	0.00228

Branching ratios of various decay modes for a 125 GeV Higgs boson.



arXiv:1307.1347

Potential Di-Higgs Decay Channels

$hh \to b\bar{b}\gamma\gamma$

Reasonable cross section ($BR \sim 0.26\%$), good background rejection wth photons.

- $hh \to b\bar{b}b\bar{b}$

  Large cross section ($BR \sim 67\%$), but large QCD backgrounds/ poor mass resolution. Important for investigating m_{hh} tail, sensitive to derivative operators.

- $hh \to b\bar{b}4\ell$

  Small cross section, but distinctive signature.

- $hh \to b\bar{b}\ell^+\ell^-(+\not{E}_T)$

  Contributions from $h \to W^+W^-, \tau^+\tau^-, \mu^+\mu^-$.

$hh \to 4\gamma$ channel produces $\sim 50(8)$ events/3 ab^{-1} at 100 (33) TeV.

Prospects at Future Colliders – Snowmass Study

ArXiv:1308.6302 estimated precision at 14, 33, and 100 TeV with a common set of kinematic cuts.
Estimated precision on λ_3 of 50%, 20%, and 8%, respectively at 3 ab^{-1} (68% C.L.).

Room for improvement:

- Slope $d(\sigma/\sigma_{\text{SM}})/d(\lambda/\lambda_{\text{SM}}) = -0.8$, didn't correct for the acceptance.
- Didn't include light jet faking photons (e.g., $b\bar{b}j\gamma$ background).
- Can improve S/B with tighter cuts on $m_{\gamma\gamma}$, leading $p_T(\gamma), p_T(b)$, looser cuts elsewhere.

Di-Higgs Production Prospects at the HL-LHC

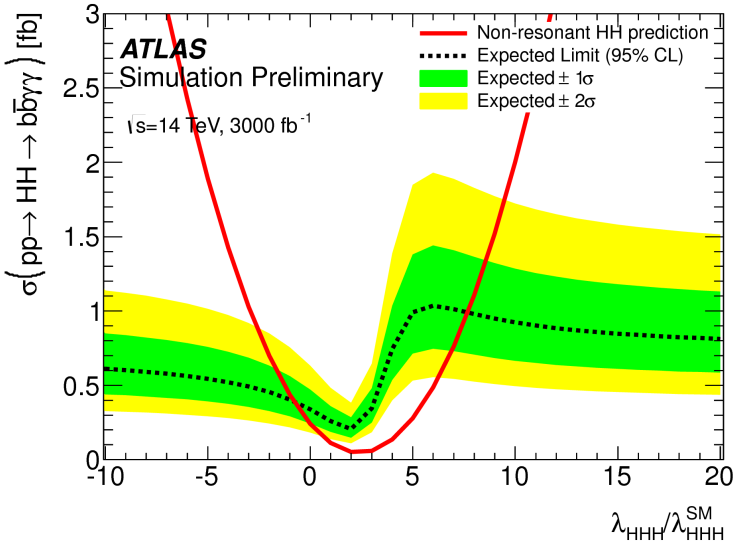
ATL-PHYS-PUB-2017-001

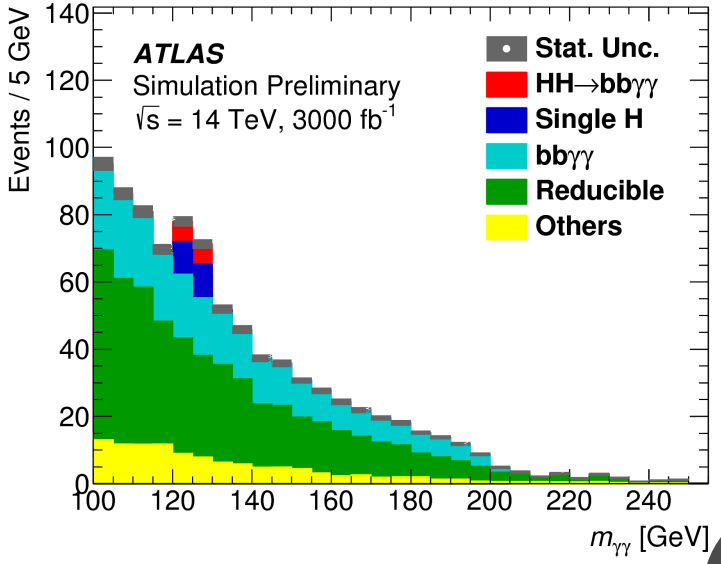
ATLAS has studied the $b\bar{b}\gamma\gamma$ channel in the context of the HL-LHC (14 TeV), assuming 3 ab^{-1} of data.

Expected significance of 1.05σ for the standard model signal.

Trilinear coupling constrainted to the range $-0.8 < \lambda_3 < 7.7$ at 95% confidence level.

$hh \to b\bar{b}b\bar{b}$ at HL-LHC estimates $0.2 < \lambda_3 < 7.0$.





Prospects at a 100 TeV Collider

A number of studies on $hh \to b\bar{b}\gamma\gamma$ at a 100 TeV collider.

ArXiv:1502.00539, 1506.03302, 1606.09408

Wide range of estimates based on detector assumptions, etc.

Estimate 1206 signal events, 2712 total background at 3 ab^{-1}.

Background dominated by $b\bar{b}j\gamma$, $jj\gamma\gamma$, $t\bar{t}h(\gamma\gamma)$

Statistical precision on λ_3 at 100 TeV improved to $\sim 5\%$ at 3 ab^{-1}.

Can we get the same improvement in precision at 33 TeV?

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Simulation of Di-Higgs Signal

The signal is generated at parton-level using `MadGraph5` with the `cteq6l1` parton density functions.

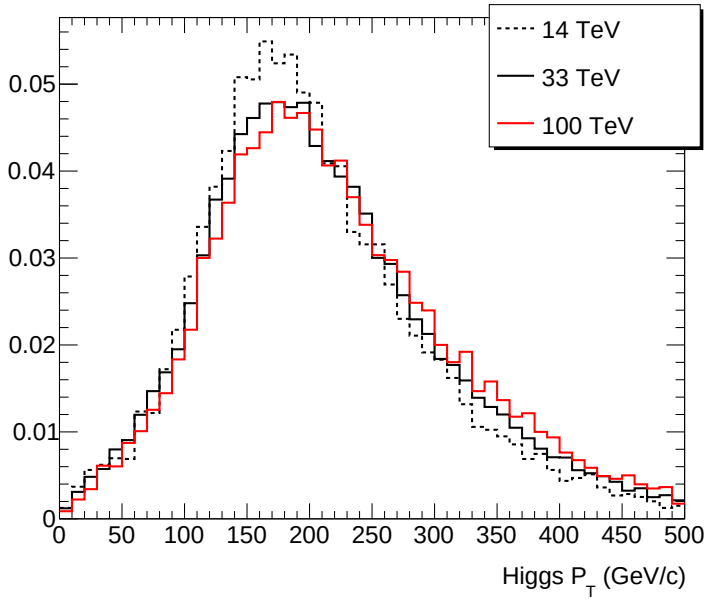
We consider only the LO (one-loop) process $gg \to h \to hh$, which can be generated directly in `MG5` (1507.00020).

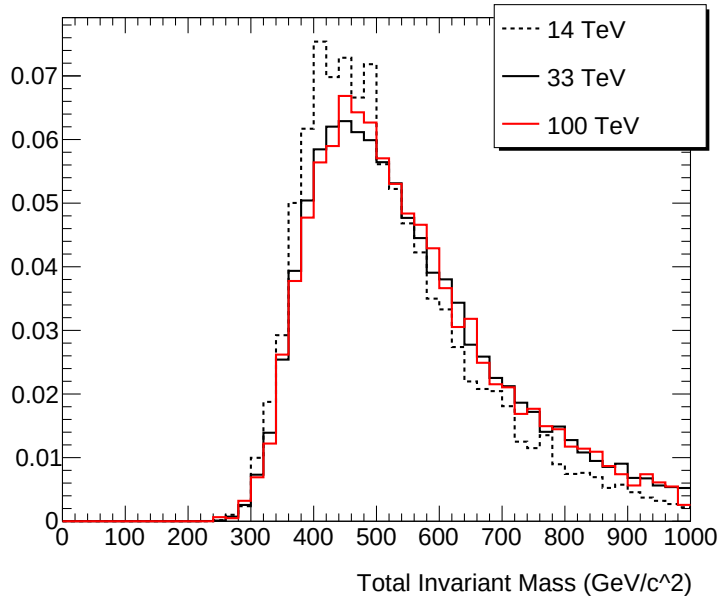
The resulting physical h particles are then decayed using Madspin.

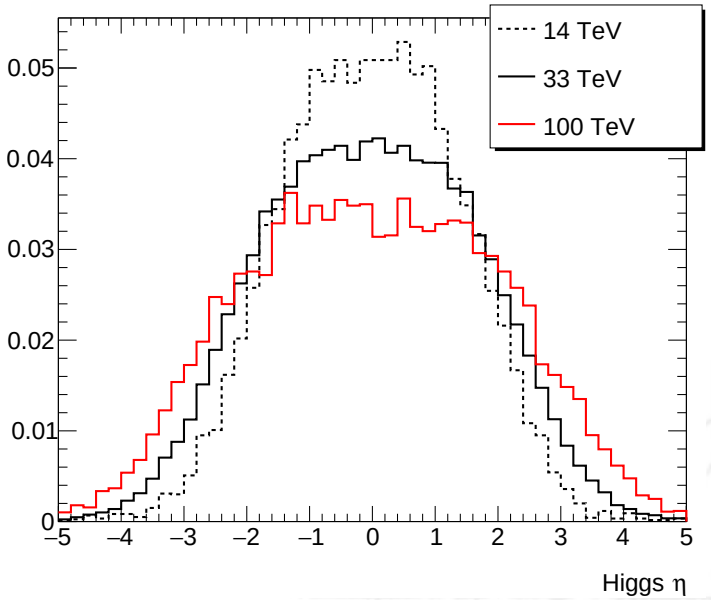
The LO signal is scaled to match the NNLL cross section in the standard model.

$\sigma \times BR = 0.70, 4.6$ fb at 33, 100 TeV.

Di-Higgs Signal Kinematics







Background Processes

The full list of background processes in our analysis is as follows:

- $t\bar{t}h(\rightarrow \gamma\gamma)$
- $b\bar{b}h(\rightarrow \gamma\gamma)$
- $b\bar{b}\gamma\gamma$
- $b\bar{b}j\gamma$
- $jj\gamma\gamma$
- $bj\gamma\gamma$

Not included in previous analysis at 33 TeV!

Additional backgrounds from $t\bar{t}\gamma$, $t\bar{t}\gamma\gamma$, $b\bar{b}jj$ shown to be insignificant.

Background Simulations

All backgrounds are generated using `MadGraph5`, with showering and hadronization effects included using `pythia6`.

To help validate our results, we follow the methods in 1606.09408:

- The $b\bar{b}\gamma\gamma$ and $bj\gamma\gamma$ samples are generated with up to one extra parton, with MLM matching used to avoid overcounting.
- Remaining backgrounds generated at LO and rescaled by an appropriate k-factor.

Detector Simulation with `Delphes3`

A custom `Delphes3` card was used to model reconstrcution with similar assumptions as in 1606.09408.

Jet flavor tagging probabilities taken flat in p_T:

$p_{b\to b} = 0.75$, $p_{c\to b} = 0.1$, $p_{j\to b} = 0.01$.

Photon reconstruction efficiency modeled as

$$\begin{cases} 95\% & \text{for } |\eta| \leq 1.5 \\ 90\%, & \text{for } 1.5 < |\eta| \leq 4 \\ 80\%, & \text{for } 4 < |\eta| \leq 6 \end{cases}$$

Also include a light-jet-to-photon mis-tagging probablity,

$$p_{j\to\gamma} = 0.01 \cdot \exp\left(-p_{T,j}/(30\ \text{GeV})\right)$$

Energy Resolution

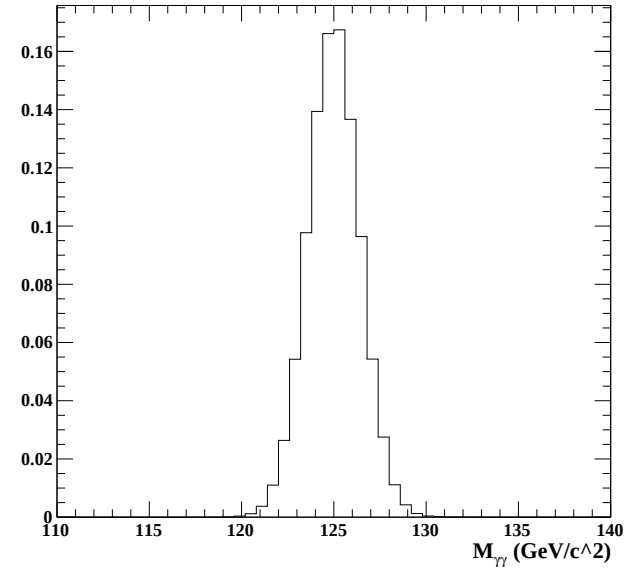
Assume calorimeter energy resolutions similar to ”Medium” performance benchmark for FCC-hh-like detector:

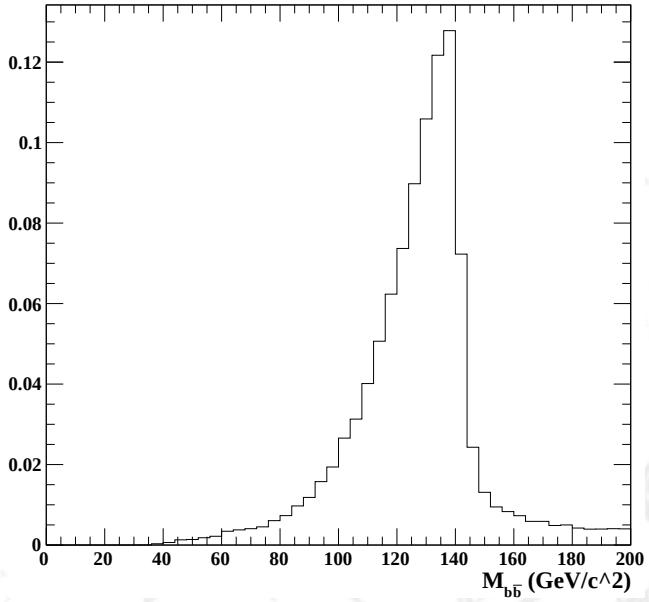
$$\Delta E = \sqrt{a^2 E^2 + b^2 E}$$

where for the E-Cal,
$a = 0.01$, $b = 0.1$.

For the H-Cal:

$$\begin{cases} a = 0.03, b = 0.05 & \text{for } |\eta| \leq 4 \\ a = 0.05, b = 1.0 & \text{for } 4 < |\eta| \leq 6 \end{cases}$$



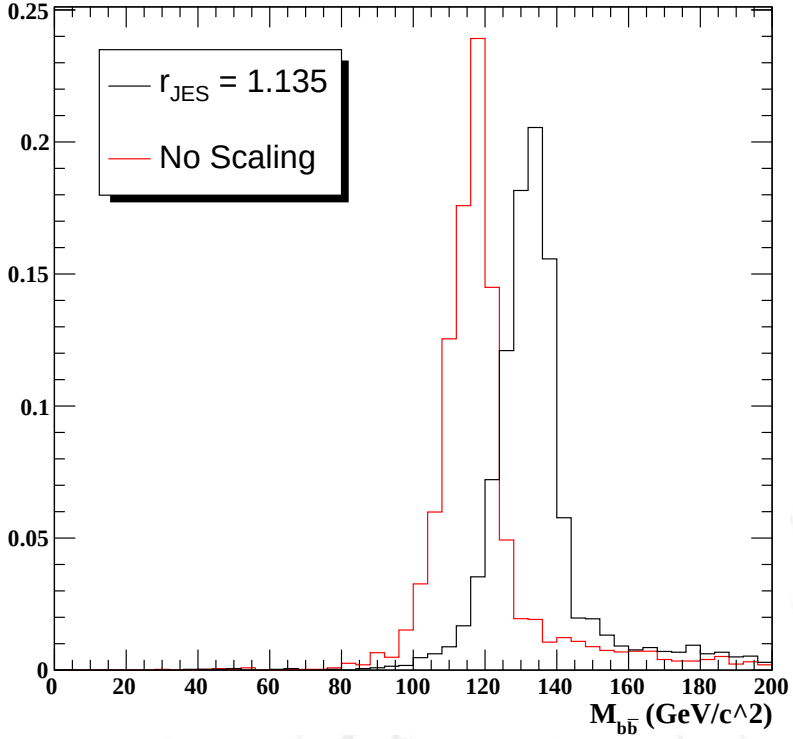


Jet Energy Scaling

Following FCC-hh benchmarks, we've assumed a high magnetic field, $B = 6$ T.

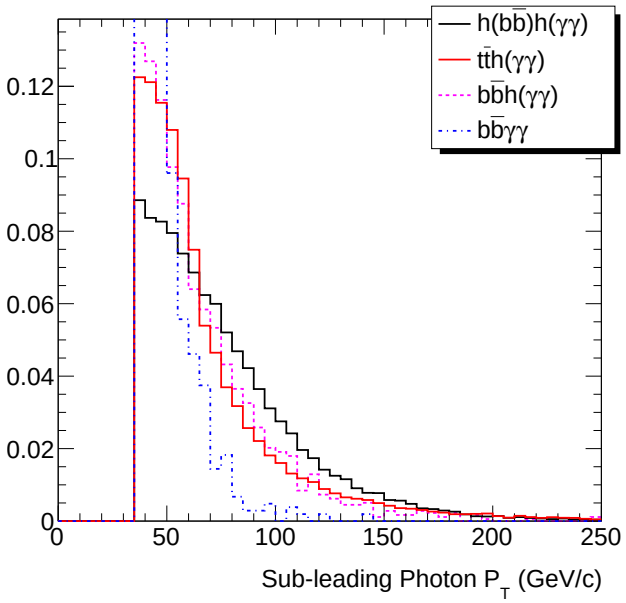
This makes it difficult to reconstruct low-p_T charged particles, which can significantly distort the jet mass distribution.

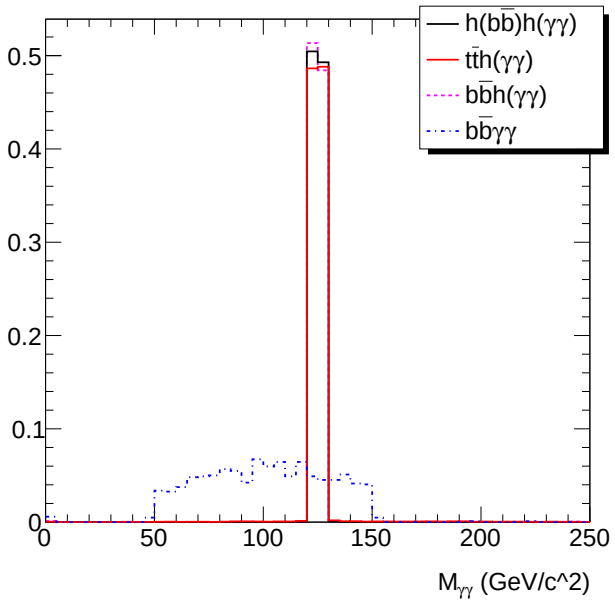
To compensate, we rescale the jet four-momentum by a factor $r_{JES} = 1.135$.

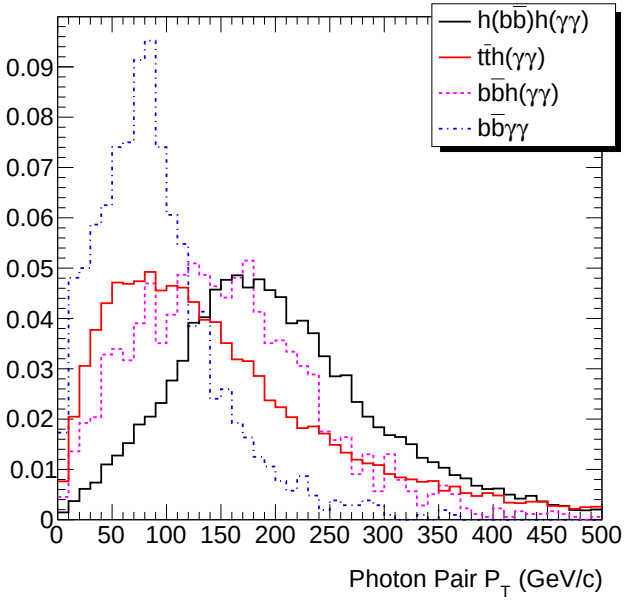


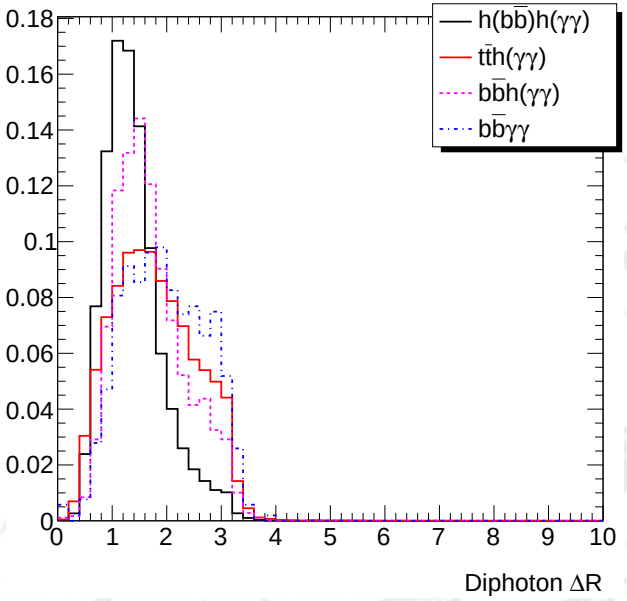
Signal and Background Kinematics

Di-Photon Distributions



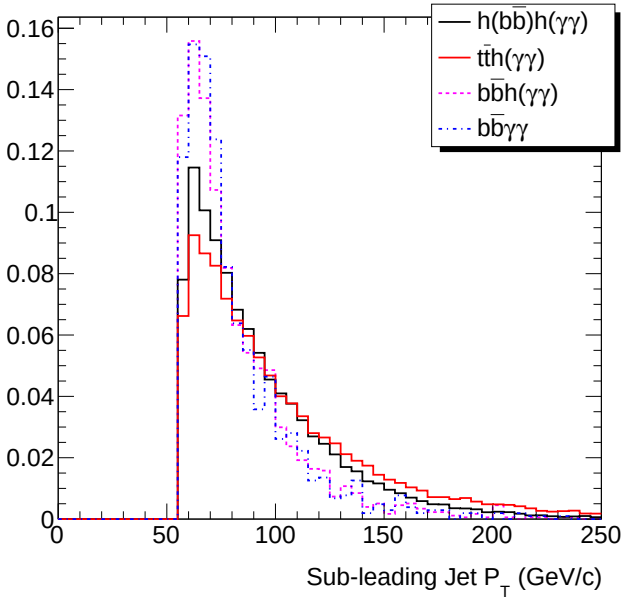


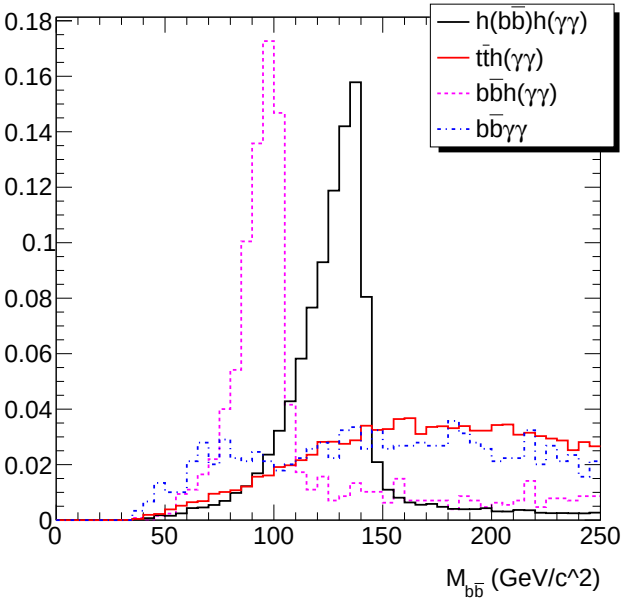


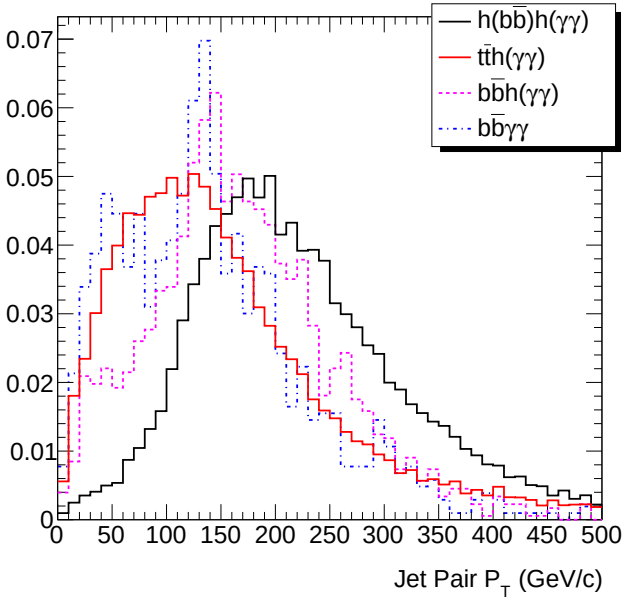


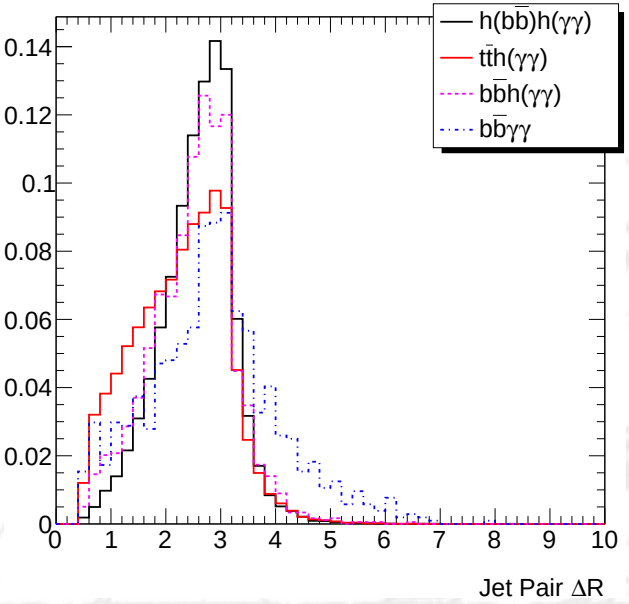
Signal and Background Kinematics

Di-Jet Distributions



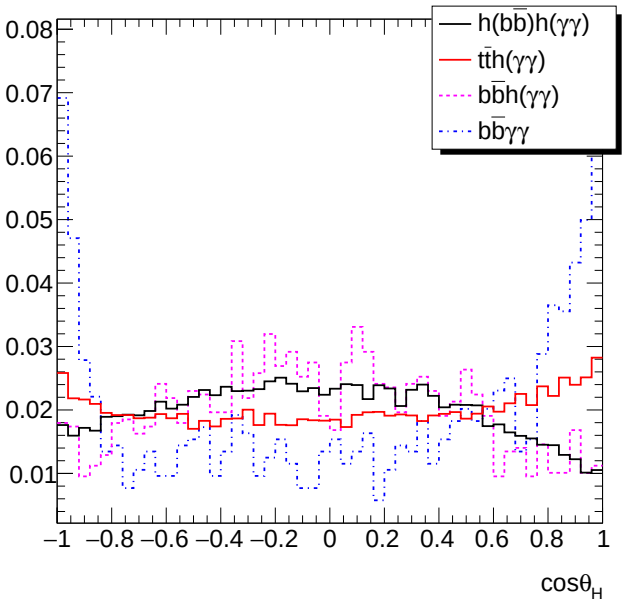


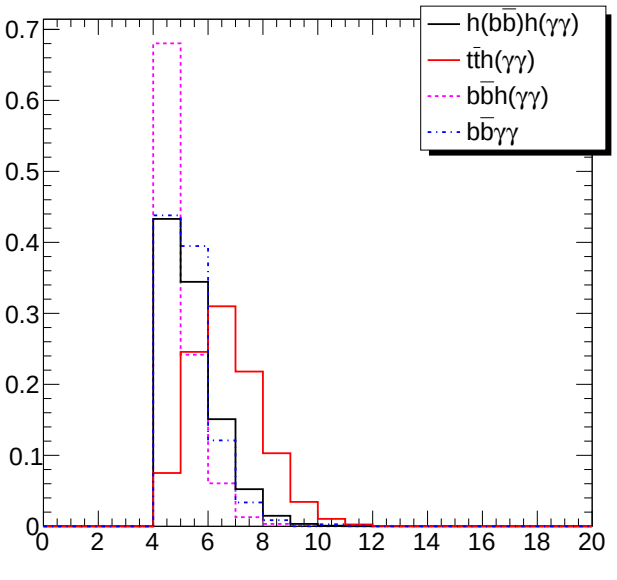


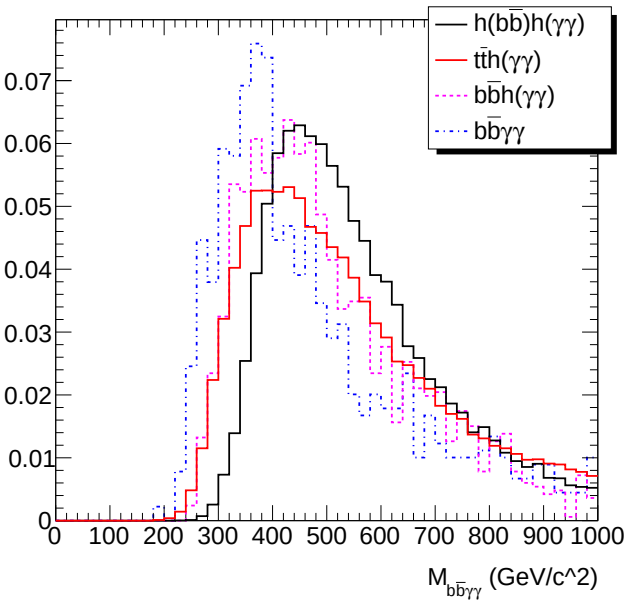


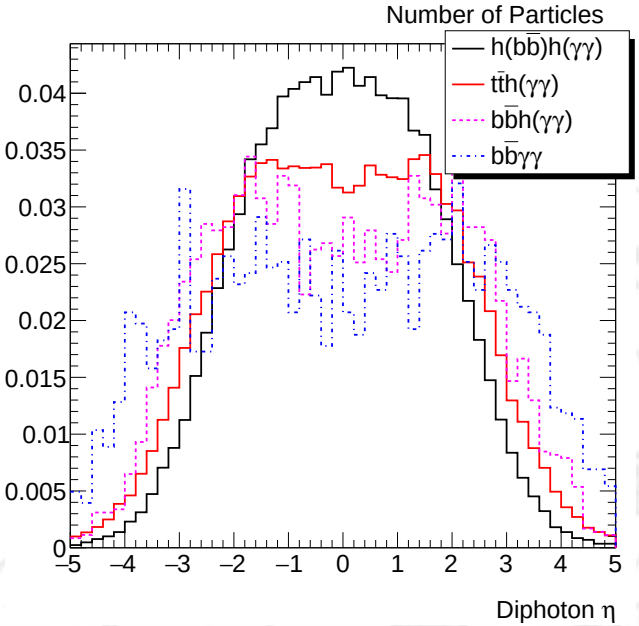
Signal and Background Kinematics

Additional Distributions









Choice of Final Cuts

Our final selection cuts were similar to those in 1606.09408:

- $|\eta_{b,\gamma}| < 4.5$
- $p_T(b_1), p_T(\gamma_1) > 60$ GeV
- $p_T(b_2), p_T(\gamma_2) > 35$ GeV
- $m_{b\bar{b}} \in [100, 150]$ GeV
- $|m_{\gamma\gamma} - m_h| < 2.5$ GeV
- $p_T(b\bar{b}), p_T(\gamma\gamma) > 100$ GeV
- $\Delta R(b\bar{b}), \Delta R(\gamma\gamma) < 3.5$
- $num_{\text{lep}}(p_T > 25 \text{ GeV}) = 0$
- $num_j(p_T > 35 \text{ GeV}, |\eta| < 6) < 6$
- $\cos\theta_h < 0.8$*

Next Steps

Lots of refining to do:

- Currently using k-factors derived at 100 TeV. These need to be recalculated.
- We need to estimate $d(\sigma/\sigma_{\text{SM}})/d(\lambda/\lambda_{\text{SM}})$ for the acceptance at 33 TeV.

We've been closely following 1606.09408 to validate our results with the 100 TeV scenario, but we should probably optimize cuts to 33 TeV case.

Several of our channels have extremely large cross sections but very low acceptances – we can improve our estimates by using an external reweighting.

Conclusions

- Measuring the Higgs trilinear coupling is an important probe of the consistency of the Standard Model and gives us insight into the electroweak phase transition.
- Precision measurements of λ_3 can be made by studying $gg \to hh \to b\bar{b}\gamma\gamma$ at high energy colliders.
- A 40% measurement of λ_3 or better seems to be possible at 33 TeV, with lots of room for improvement.

Backup: Comparison with Snowmass Study

Comparison of cuts:

Our Cuts:

- $|\eta_{b,\gamma}| < 4.5$
- $p_T(b_1), p_T(\gamma_1) > 60$ GeV
- $p_T(b_2), p_T(\gamma_2) > 35$ GeV
- $M_{b\bar{b}} \in [100, 150]$ GeV
- $|m_{\gamma\gamma} - m_h| < 2.5$ GeV
- $p_T(b\bar{b}), p_T(\gamma\gamma) > 100$ GeV
- $\Delta R(b\bar{b}), \Delta R(\gamma\gamma) < 3.5$
- $n_{leps}(p_T > 25 \text{ GeV}) = 0$
- $n_{jets}(p_T > 35 \text{ GeV}, |\eta| < 6) < 6$
- $\cos\theta_h < 0.8$

Snowmass Cuts:

- $p_T(b, \gamma) > 35$ GeV
- $|\eta(b, \gamma)| < 2.5$
- $|m_{\gamma\gamma} - m_h| < 5$ GeV
- $m_{b\bar{b}} \in [85, 135]$ GeV
- $\Delta R_{\gamma\gamma} < 2.5, \Delta R_{b\bar{b}} < 2.0$
- $|\eta_{\gamma\gamma}|, |\eta_{b\bar{b}}| < 2.0$
- $p_{T,\gamma\gamma}, p_{T,b\bar{b}} > 100$ GeV
- $m_{b\bar{b}\gamma\gamma} > 300$ GeV
- $\cos\theta_h < 0.8$
- $\sum \left(n_{jets} + n_\gamma + n_{lep} + n_{met}\right) < 7$

Backup: Strong EWPT in a "Nightmare Scenario"

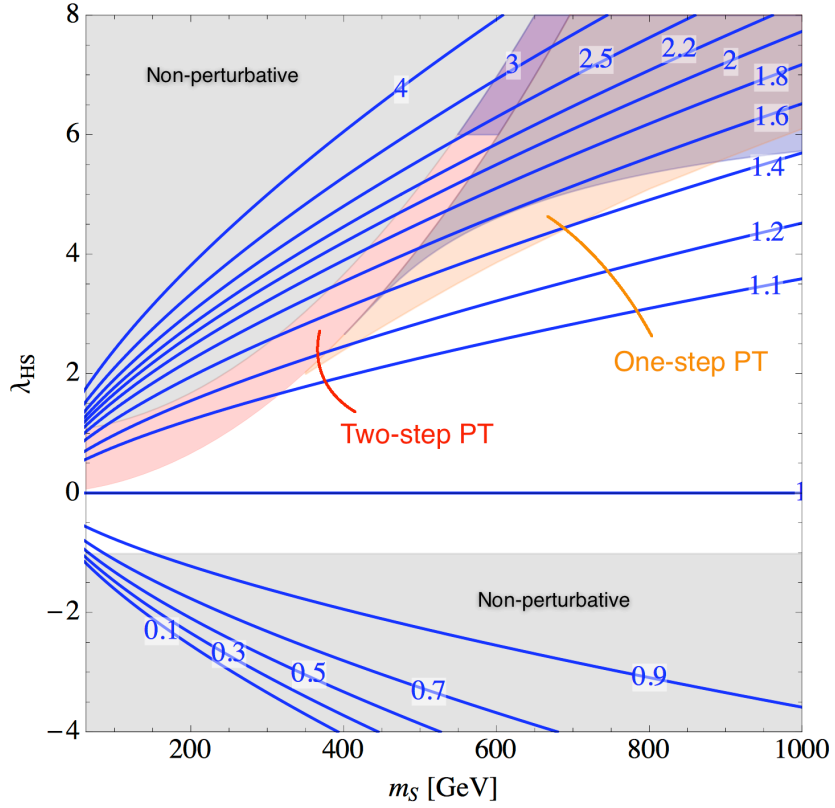
arXiv:1409.0005

"Nightmare scenario" includes an extra singlet scalar S with $\mathbb{Z}_2$ symmetry to avoid singlet-Higgs mixing:

$$V_0 = -\frac{1}{2}\mu^2 h^2 + \frac{1}{4}\lambda h^4 + \frac{1}{2}\mu_S^2 S^2 + \frac{1}{2}\lambda_{HS} h^2 S^2 + \frac{1}{4}\lambda_S S^4$$

Leads to:

- One-step transition with $\mu_S^2 > 0$
- or Two-step transition with $\mu_S^2 < 0$

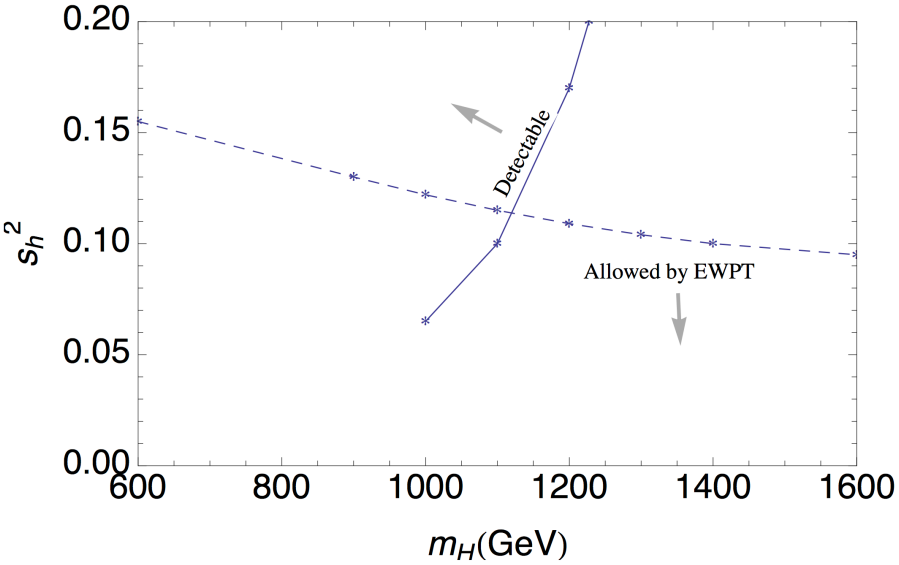


Backup: Maximal Deviations in H+S Model

arXiv:1305.6397

For a mixed-in singlet, mass eigenstates become

$$\phi_{SM} = \cos\theta_h h + \sin\theta_h S$$
$$\phi_S = -\sin\theta_h h + \cos\theta_h S$$



The Higgs cubic interaction becomes

$$g_{hhh} = \frac{m_h^2}{2v}\left(\cos^3\theta_h - \frac{v}{v_s}\sin^3\theta_h\right)$$

Bounds from detection of S and electroweak precision tests limit $\sin^2\theta_h \lesssim 0.12$, which translates to $\Delta\lambda_3 \approx -18\%$.

Backup: $hh \to bbbb$ Studies

ATLAS has published (ATL-PHYS-PUB-2016-024) expected limits on λ_3 from extrapolating their 2016 data set to the HL-LHC with 3 ab^{-1} of data.

Expect λ_3 to be constrained to $0.2 < \lambda_3 < 7.0$, with only statistical uncertainties. However, this range expands to $-3.5 < \lambda_3 < 11$ when projecting current systematics.

A 100 TeV study used MVA techniques on 3 categories of events, based on the size of the jets (R) and the number of tags.

Find S/B smaller at 100 TeV than at 14 TeV, due to rapid growth of the QCD $4b$ background.

	$\delta_{\text{sys}}\sigma = 25\%$	$\delta_{\text{sys}}\sigma = 100\%$
Boosted	$\lambda_3 \in [-0.1, 2.2]$	$\lambda_3 \in [-1.5, > 9]$
Intermediate	$\lambda_3 \in [-0.7, 1.6]$	$\lambda_3 \in [-0.4, > 9]$
Resolved	$\lambda_3 \in [0.9, 1.5]$	$\lambda_3 \in [-0.1, 7]$